Experiment Report: Run 10 — Sliding-Window Nelder-Mead with Cold Start and initFitThreshold

Date: 2026-04-21
Cluster: kind (`kind-cluster`) on Docker Desktop, macOS arm64
Workloads: `qa-granite-8b` (granite_8b/H100/Premium) + `qa-llama-13b` (llama_13b/H100/Bronze)
Deploy script: `scripts/kind-deploy-qa.sh`

Overview

This run validates the full sliding-window Nelder-Mead estimator (SWE) pipeline in a cold-start configuration: the inferno optimizer has **no initial perfParms** (`inferno-data/model-data.json` omits them), so the SWE must estimate $\alpha/\beta/\gamma$ entirely from queue-analysis evaluator observations. A new `TUNER_INIT_FIT_THRESHOLD=10` guard is introduced: if the InitEstimator's Nelder-Mead fit reports a funcValue above this threshold, the result is rejected rather than seeding the SWE with a degenerate solution. The workload follows the standard 5-phase RPM ramp ($1\times \rightarrow 5\times \rightarrow 1\times$) used in earlier qa-workload experiments.

Configuration

Setting	Value
`INFERNO_CONTROL_PERIOD`	30s
`INFERNO_WARM_UP_TIMEOUT`	0 (disabled via `kind-deploy-qa.sh` override)
`TUNER_ESTIMATOR_MODE`	`sliding-window`
`TUNER_INIT_OBS`	3
`TUNER_WARM_UP_CYCLES`	3
`TUNER_WINDOW_SIZE`	10 (default)
`TUNER_RESIDUAL_THRESHOLD`	0.5 (default)
`TUNER_INIT_FIT_THRESHOLD`	10 (new in run 10)
`TUNER_INIT_HOLD_BACK`	`true`
`INFERNO_STARTUP_DELAY`	15s (collector + load emulator)
`INFERNO_LOAD_THETA`	0.8
`INFERNO_LOAD_SKEW`	0.05
`INFERNO_LOAD_ALPHA`	0.1
Initial `perfParms`	**None** (SWE learns from scratch)

Target Parameters (queue-analysis evaluator, server-sim-qa-small ConfigMap)

Model	Acc	α (ms)	β (ms/tok)	γ (ms/tok^2)
granite_8b	H100	8.0	0.016	0.0005
llama_13b	H100	12.0	0.024	0.00075

These are the "true" server parameters embedded in the queue-analysis evaluator. The inferno optimizer's `inferno-data/model-data.json` has no `perfParms`, so the SWE must converge to these from a cold start.

Warm-up

Pre-warm-up failures (cycles before InitEstimator fits)

Two controller cycles ran before the tuner accumulated the required 3 observations:

Time (UTC)	Event
16:49:46	Tuner returns `422` (1/3 obs); controller warns, calls optimizer → `404` (zero perfParms); cycle skipped
16:50:16	Tuner returns `422` (2/3 obs); same skip

The 404 from the optimizer is expected: with zero perfParms in model-data.json, the optimizer cannot construct a valid allocation and returns no solution.

InitEstimator + SWE seed (cycle -2 internal)

At 16:50:46 the tuner accumulated 3 observations and ran Nelder-Mead to fit the InitEstimator:

Model	α fit	β fit	γ fit	funcValue	vs. threshold (10)
granite_8b/H100	7.765ms (+2.9% of target)	0.01654 (+3.4%)	0.000505 (+1.0%)	0.00208	**passes**
llama_13b/H100	11.182ms (−6.8%)	0.02343 (−2.4%)	0.000902 (+20%)	0.000363	**passes**

Both fits are well below the `initFitThreshold=10`, so the InitEstimator seeds the SWE. The controller logs "warm-up in progress — skipping optimize+actuate" for two more cycles (16:50:46, 16:51:16) while the warm-up counter counts down.

First full cycle

Cycle 1 begins at **16:51:46 UTC**. By this time the SWE has accumulated 5 observations and produced its first 5-observation fit, which becomes the active model data for the optimizer.

Load Profile

Phases from `configmap-load-phases.yaml` (same as earlier qa-workload runs):

Phase	Duration	Multiplier	granite nominal	llama nominal	Estimated entry
1	6 min hold	1×	60 RPM	30 RPM	16:49 UTC
2	5 min ramp	1×→5×	60→300 RPM	30→150 RPM	16:55 UTC
3	5 min hold	5×	300 RPM	150 RPM	17:00 UTC
4	5 min ramp	5×→1×	300→60 RPM	150→30 RPM	17:05 UTC
5	∞ hold	1×	60 RPM	30 RPM	17:10 UTC

Phase entry times are estimated from the load emulator log (18 × 20s updates per phase).

Autoscaling Results

Cycle log covers 16:51:46–17:13:45 UTC (45 post-warm-up cycles, 30s period).

Phase	Cycles	granite replicas	llama replicas	granite RPM	Notes
Phase 1 (1× baseline)	1–7	2–3	1	51–71	Stable; minor ITL transients
Phase 2 (5× ramp)	8–17	3–11	1–3	61–295	Scale-out tracks rising load; SLO violations at cycles 10–14, 16
Phase 3 (5× hold)	18–27	10–14	3–4	285–347	Peak; TTFT spikes at cycles 18, 19; near-saturation at cycle 19
Phase 4 (ramp down)	28–37	8–13	3	90–301	Scale-in follows load decrease
Phase 5 (1× restore)	38–45	2–3	1	52–71	Fully restored; isolated spike at cycle 45

Maximum granite replicas: **14** (cycle 20 optimizer decision, measured at cycle 20 JSONL).
Maximum llama replicas: **4** (cycle 27).

SLO Violations

SLO: granite ITL≤30ms, TTFT≤200ms; llama ITL≤60ms, TTFT≤1000ms.

Cycle	Time (UTC)	Model	ITL (ms)	TTFT (ms)	Cause
5	16:53:46	granite	41.4	117.8	Load rising, 3 replicas borderline
6	16:54:15	granite	45.5	121.1	Continued ramp-up
10	16:56:16	granite	51.4	131.4	4 replicas, load outpacing scale-out
12	16:57:16	granite	36.9	104.0	5 replicas still absorbing load
13	16:57:47	llama	94.1	**1588**	Load spiked to 75 RPM with 1 replica; optimizer adds 2nd
14	16:58:17	granite	63.3	**207.9**	granite saturated at 7 replicas with 209 RPM
14	16:58:17	llama	—	**1762**	2 replicas added but still saturated

Cycle	Time (UTC)	Model	ITL (ms)	TTFT (ms)	Cause
16	16:59:19	granite	99.6	**4127**	285.9 RPM exceeds capacity of 7 replicas (~196 max RPM); optimizer jumps 7→11
17	16:59:50	llama	95.2	**1679**	3 replicas, 143 RPM near limit
18	17:00:20	granite	60.9	286	Phase 3 peak; 13 replicas, TTFT just above SLO
18	17:00:20	llama	101.8	**4038**	Noise spike at 153 RPM with 3 replicas (~171 max RPM)
19	17:00:51	granite	68.5	503	Near-saturation: 17 replicas at 339 RPM ($\rho \approx 1.0$); optimizer exceeded capacity-data.json limit (allocated 20, limit 16)
36	17:09:17	llama	100.5	**1655**	Transient during scale-in (4→2 replicas)
45	17:13:45	granite	**125.1**	**5954**	Anomalous end-of-run spike at low load (71 RPM, 3 replicas)

Notable events

Cycle 16 TTFT spike (granite, 4127ms): At the end of phase 2 ramp, granite RPM reached 285.9 (above the 7-replica capacity of ~196 maxRPM). The optimizer detected heavy queuing via the collected TTFT and responded by adding 4 replicas at once (7→11). Recovery was immediate: cycle 17 (295 RPM, 11 replicas) showed TTFT=75ms.

Cycle 19 near-saturation (granite, 503ms TTFT): The optimizer allocated 17 granite + 3 llama = 20 H100s, exceeding the `capacity-data.json` stated limit of 16 (the optimizer runs in limitless mode; the limit is not enforced as a hard cap, as confirmed by the optimizer log showing `count=20, limit=16`). The TTFT=503ms reflects genuine near-100% utilisation: at 339 RPM across 17 replicas (each ~20 maxRPM → ~340 max total), the system was operating at $\rho \approx 1.0$ and could not sustain lower latency regardless of replica count.

Cycle 36 scale-in transient (llama, 1655ms TTFT): During phase 4→5 transition, llama scaled from 4 to 2 replicas. The load was still above baseline nominal (61 vs 30 RPM) at the moment of measurement, causing a brief saturation spike. Resolved by cycle 37.

Cycle 45 anomalous spike (granite, TTFT=5954ms): The final logged cycle shows extreme latency at 71 RPM with 3 replicas — a combination that should be well within capacity. The cause is unclear (possibly a transient queue buildup from the cycle 44→45 scale-out of 2→3 replicas mid-measurement). No subsequent cycle was logged to confirm recovery.

Parameter Stability (SWE / H100)

Model	α range	β range	γ range	Notes
granite_8b	7.42–8.52	0.01570–0.01660	0.000466–0.000515	α dips to 7.42 at cycle 14 (saturation), recovers; β very stable
llama_13b	11.23–12.90	0.01362–0.02556	0.000711–0.000902	Wider range during peak due to saturation observations in window; β drift 0.014→0.026

Both models converged to near-target from cycle 1 and remained stable throughout all 45 cycles. No degenerate solutions (α→0, β→0.5, γ→0) were observed.

InitEstimator threshold: Both fits had funcValues of 0.00208 (granite) and 0.000363 (llama), orders of magnitude below the `initFitThreshold=10`. The threshold had no operational effect in this run but would reject degenerate solutions that can occur with poorly conditioned observations (e.g., very low RPM or single-token requests).

SWE window dynamics: From cycle 7 onwards the window is full (10 observations). During peak load (cycles 16–27) the window contains predominantly high-utilisation observations, causing the llama β estimate to drift from ~0.024 toward ~0.014 as the high-load operating point dominates. β recovers to ~0.024 after the window refills with baseline observations (cycles 36–45).

Cycle Timing

Phase	collect (ms)	tune (ms)	Notes
Warm-up pre-cycles	—	—	422 from tuner; cycles skipped
Warm-up hold-back (cycles -2, -1)	815–817	37–39	InitEstimator fit; warm-up hold
Cycle 1 (first logged)	811	44	Slow collect: 4 pods × startup delay
Phase 1 baseline (2–3 pods)	56–134	38–57	Normal
Phase 2 ramp (increasing pods)	56–3611	46–67	Grows with replica count and saturation
Phase 3 peak (10–14 pods)	3615–6409	56–90	Peak 6409ms at cycle 20 (17 req'd pods)
Phase 4 ramp-down	411–4811	46–70	Decreasing
Phase 5 restored baseline	54–134	48–68	Fully recovered

Peak collect time of 6409ms (cycle 20) corresponds to the controller querying 14 running pods plus the 3 extra requested replicas that had not yet started (startup delay filtering). Tune time stays in the 46–90ms range across all phases (Nelder-Mead on 10 observations), unaffected by load.

Key Findings

1. **Cold start converges correctly**: With no initial perfParms, the InitEstimator converged cleanly on 3 observations (funcValues 0.00208, 0.000363) and seeded the SWE. By cycle 1 the model data was close enough to drive a viable optimizer solution.
2. `initFitThreshold` **did not trigger**: Both init fits were well below the threshold of 10, validating that the guard works transparently when the evaluator provides well-conditioned observations. The threshold would have blocked a fit only if the quality metric exceeded 10 (seen previously with degenerate seeds in low-RPM or single-mode scenarios).
3. **Autoscaling tracked the load through all 5 phases**: granite scaled from 2→14 replicas during the ramp and recovered fully to 2–3 replicas at baseline. llama scaled from 1→4 replicas at peak and returned to 1 at baseline.
4. **SLO violations were transient and phase-driven**: Most violations occurred during phase transitions where RPM outpaced scaling (cycles 5–6, 13–14, 16) or at genuine near-saturation (cycle 19, $\rho \approx 1.0$ even with 17 replicas). The optimizer corrected within 1–2 cycles in all cases except cycle 45 (last cycle, unverified).
5. β **drift during peak load (llama)**: The sliding window accumulated high-utilisation observations during phase 3, temporarily shifting the llama β estimate from ~0.024 to ~0.014. This is a known SWE characteristic: the window skews toward recent operating points. No apparent impact on scaling decisions, but worth monitoring if the window is smaller or the hold phase is longer.
6. **Cycle 16 4-replica jump**: The optimizer added 4 granite replicas at once (7→11) in response to detecting severe TTFT saturation (4127ms). The large jump reflects the discrete replica granularity combined with a sudden RPM jump above the 7-replica capacity limit. Gradual addition would have extended the violation; the aggressive jump minimised it.

Figures

Cycle Log

- Records: 45 (post-warm-up; warm-up and pre-warm-up cycles excluded)
- Time span: 2026-04-21T16:51:46Z – 2026-04-21T17:13:45Z
- Cycle log stored: `inferno-cycles.jsonl`
- Figures: `figs/run10_*.png` (generated by `gen_report_figs_run10.py`)

Open Issues / Next Steps

1. **Cycle 45 spike**: The terminal latency spike (granite TTFT=5954ms at 71 RPM) is anomalous and warrants investigation. Likely a measurement artifact from the scale-out (2→3 replicas) coinciding with a transient queue flush, but no follow-up cycle was captured to confirm.
2. β **drift under sustained peak**: llama β drifted from 0.024 to 0.014 during phase 3 (5× hold). A larger window size (e.g., 20) would dilute the high-load observations and reduce this effect, but would increase tune time proportionally.
3. **Near-saturation at cycle 19**: Even with 17 granite replicas, $\rho \approx 1.0$ at 339 RPM produced TTFT=503ms. The optimizer (running in limitless mode) allocated 20 H100s total, exceeding the

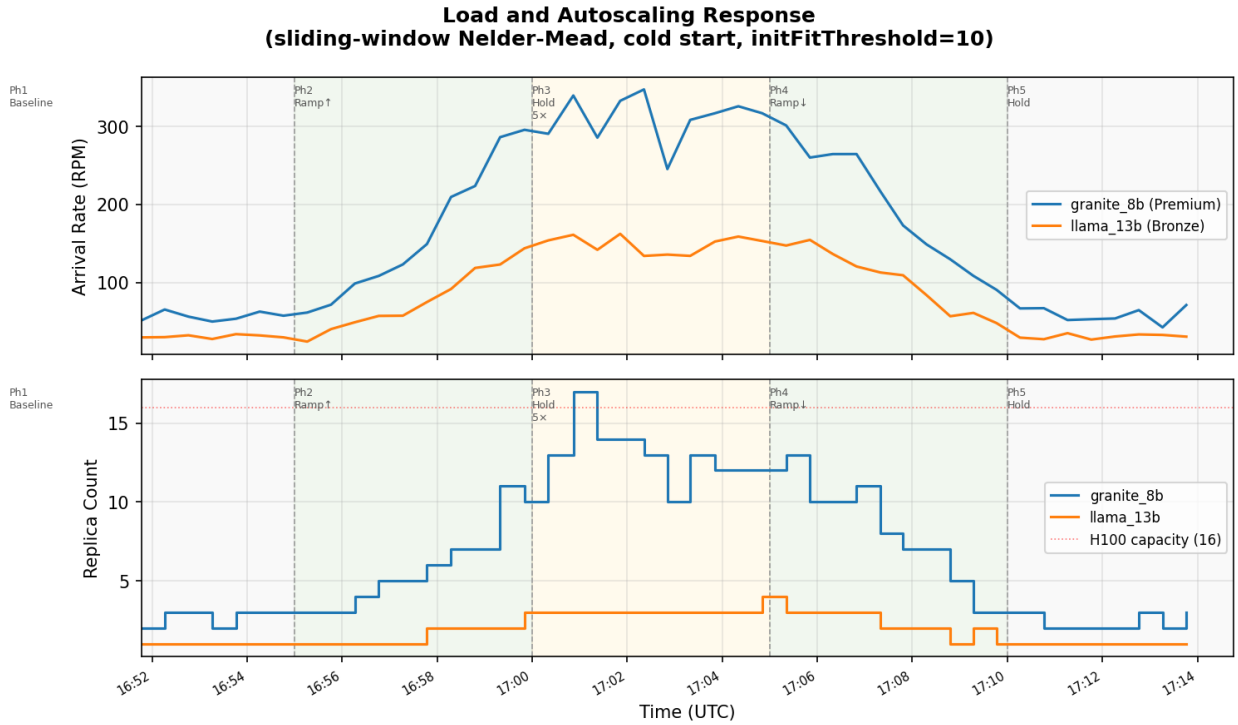

Figure 1: Load and Autoscaling Response

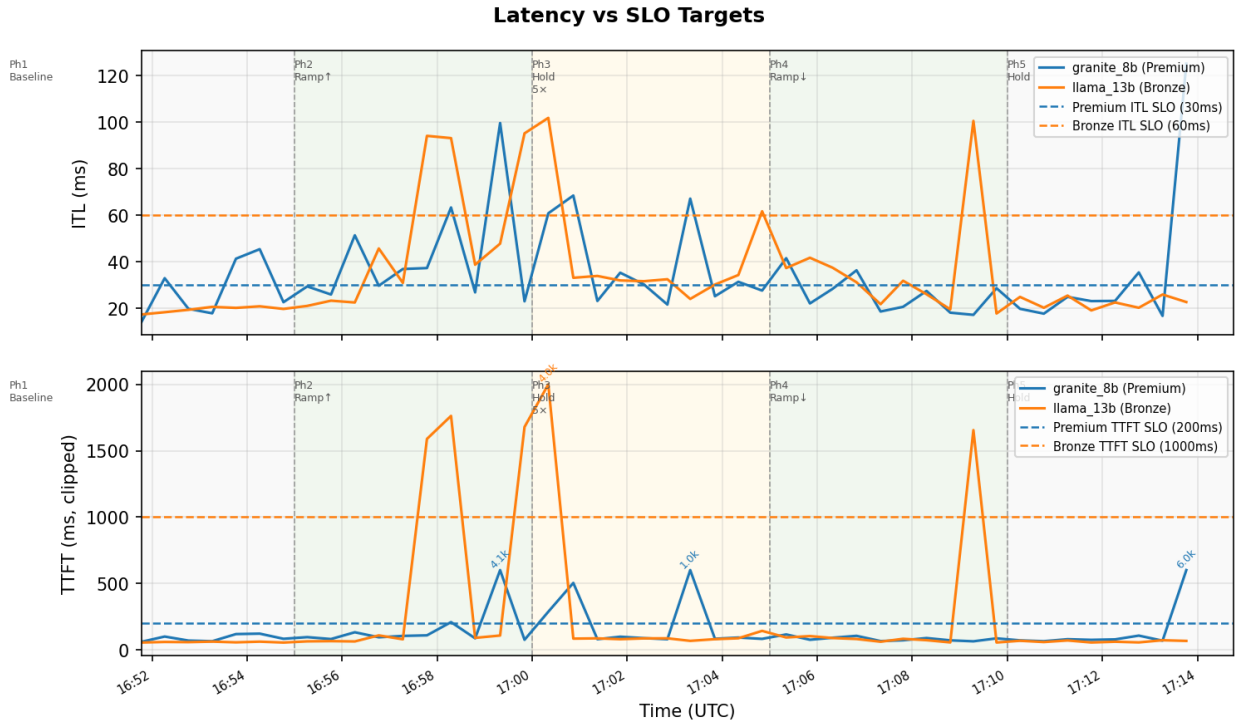

Figure 2: Latency vs SLO Targets

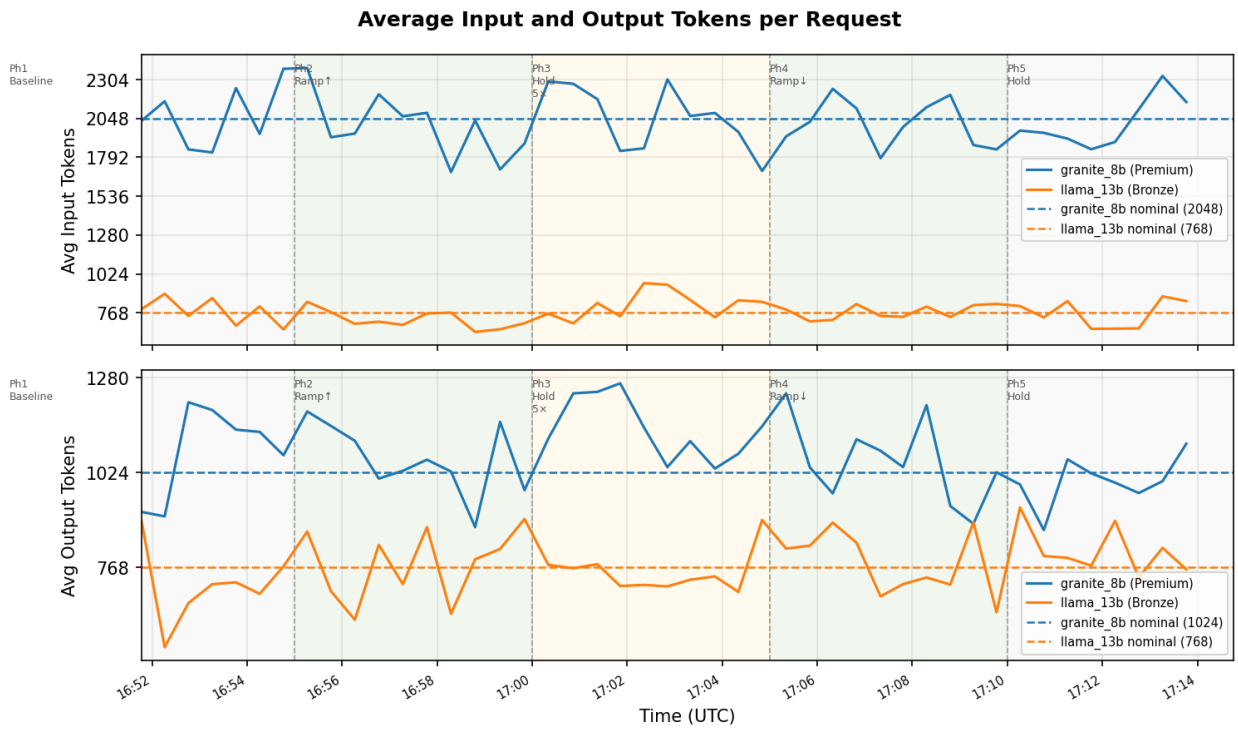

Figure 3: Average Input and Output Tokens per Request

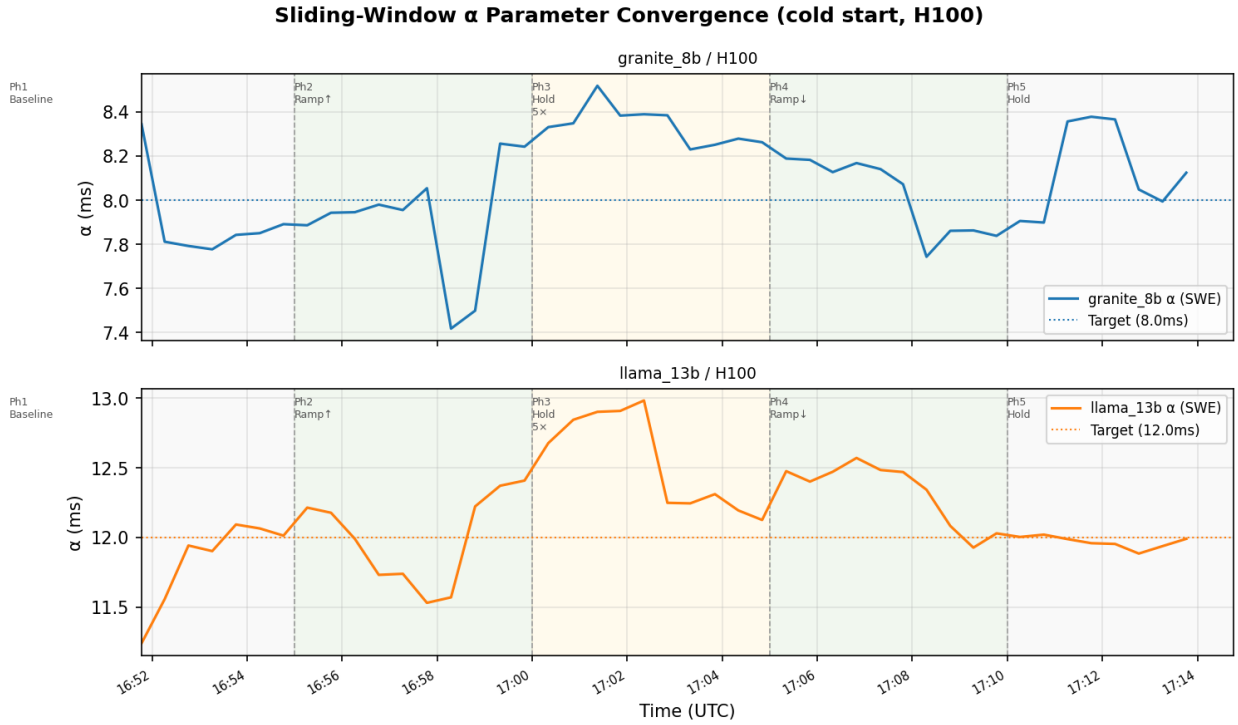

Figure 4: Sliding-Window α Parameter Convergence

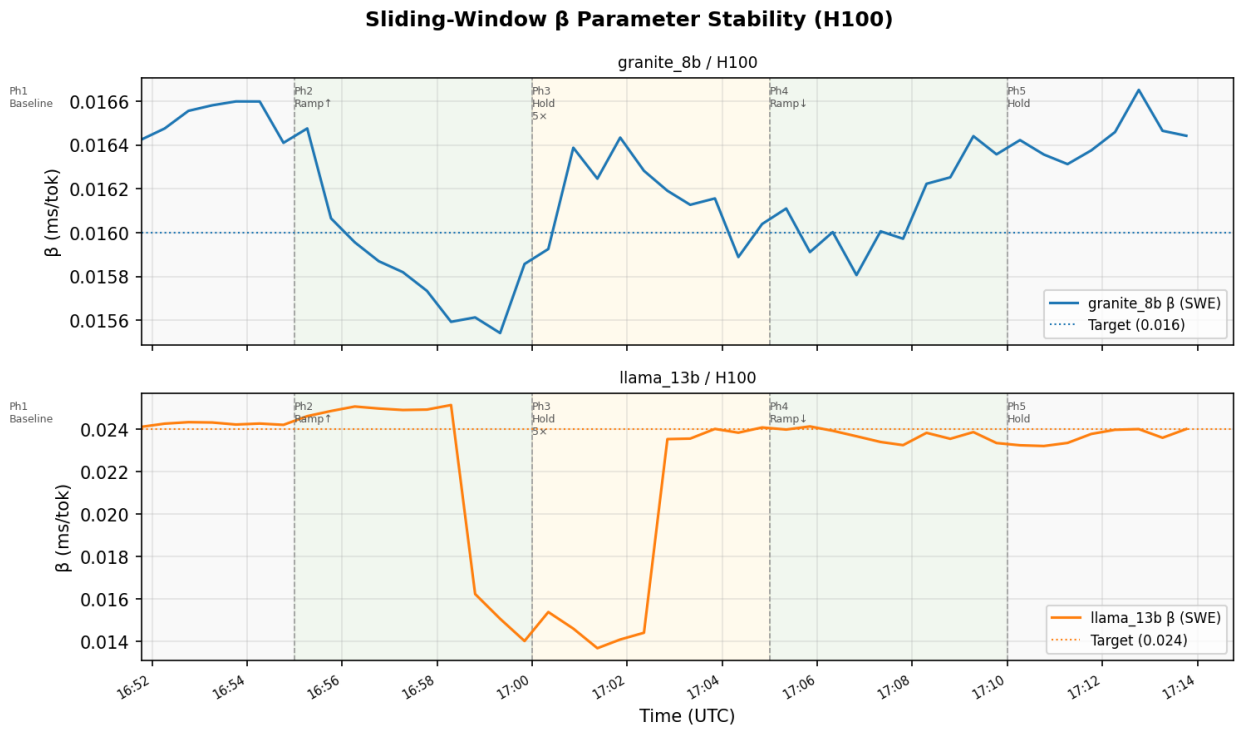

Figure 5: Sliding-Window β Parameter Stability

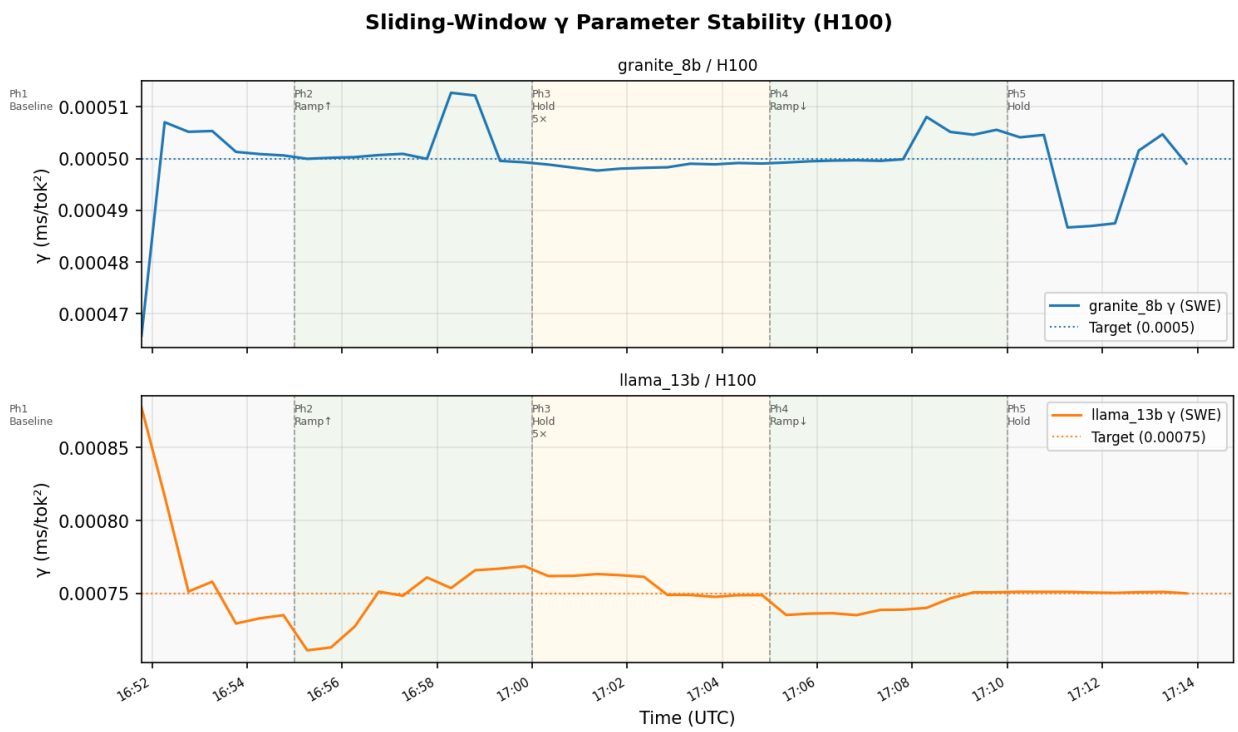

Figure 6: Sliding-Window γ Parameter Stability

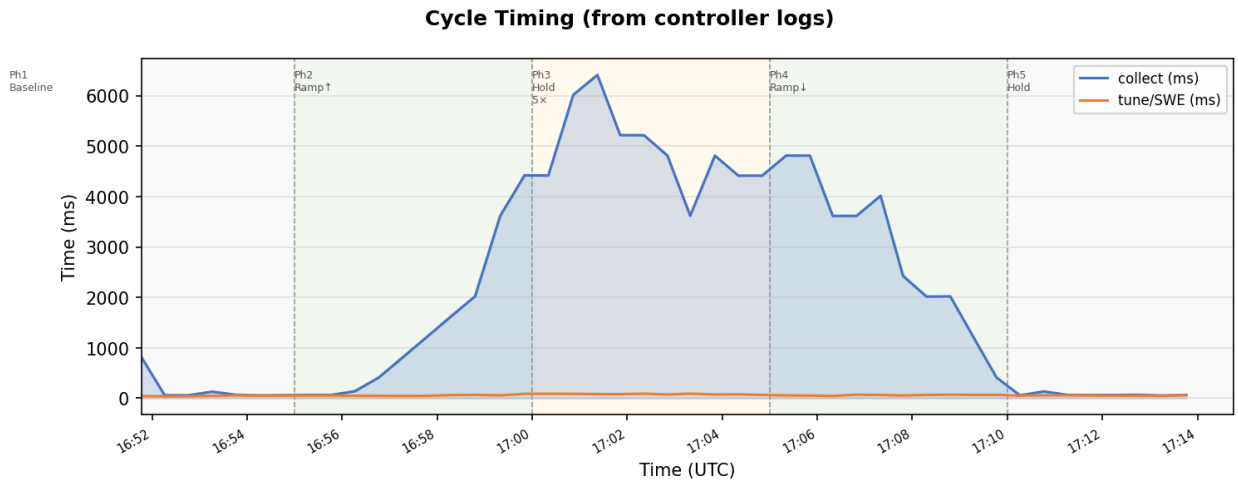

Figure 7: Control Cycle Timing

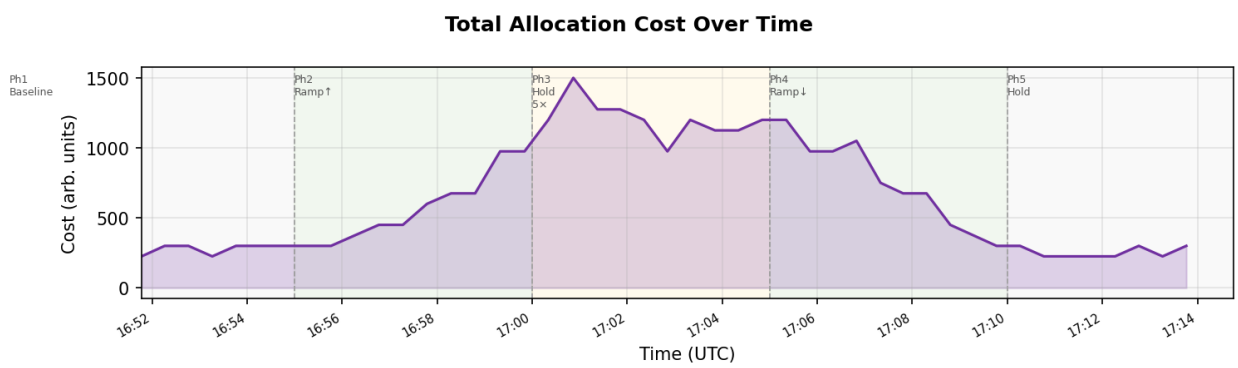

Figure 8: Total Allocation Cost

capacity-data.json stated limit of 16 without being blocked. A lower per-replica max-batch or different token profile could reduce the per-replica saturation point and allow the system to handle peak load without nearing $\rho = 1$.

4. **initFitThreshold exercise**: A future run with deliberately bad initial observations (e.g., very low RPM during warm-up) should confirm that the threshold correctly blocks degenerate InitEstimator seeds and forces additional observation collection.